

Bancor Plus: A Reserve-Curve Algorithm for Enterprises and Teams in the AI Era

Connecting AI contribution rewards, organizational budgets, adaptive reserves, and auditable settlement

Technical research paper | Test-data snapshot | English edition

Abstract

Bancor introduced an early systematic model for replacing synchronous order matching with reserve assets and continuously computed prices. This paper traces the origin of that idea, explains constant reserve ratios, transaction-size-dependent execution, and two-reserve conversion, and then applies the current Bancor Plus implementation to enterprises and teams in the AI era. Bancor Plus combines an AIC/USD exact-input reserve curve with versioned quotes, signed authorization, atomic ledgers, a participation fund, and an adaptive controller for one-sided USD reserve additions. The controller jointly applies a target share, a balance buffer, minimum and maximum rates, and a per-step reserve-growth cap. A reserve addition changes marginal price and sell-side depth but creates no trade, fee, volume, or AIC flow. The architecture links internal rewards for AI compute, model services, data contributions, and project collaboration to computable organizational liquidity. On top of this market layer, the paper proposes a profit-distribution layer funded only by settled AI-business revenue and trading fees. After costs, liabilities, and risk reserves, distributable profit is divided among liquidity, teams, users, and retained earnings, then assigned through verifiable contribution scores. The paper defines Bancor as a reserve algorithm independent of its execution medium, and discusses database settlement, on-chain contracts, and hybrid deployment as a separate execution layer. It derives the formulas, documents the test-data policy snapshot, works through numerical examples, and evaluates convergence, observability, auditability, and limitations.

Keywords: Bancor; enterprise AI; team incentives; internal value exchange; reserve curve; adaptive liquidity; auditable ledger

Bancor Plus is the name used in this system for its enhanced reserve-curve market. It does not denote an official Bancor version. USD is an internal accounting unit and carries no promise of fiat redemption.

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1. Introduction

The core liquidity problem is not whether an asset can be recorded, but whether a holder can obtain a predictable conversion when no opposite order appears at the same time. An order book depends on buyers and sellers meeting in price and time. A reserve-curve market instead lets a trader face asset reserves and a public state-transition formula. The mechanism does not remove risk; it makes part of liquidity provision computable and verifiable.

Enterprises and teams in the AI era face two linked problems: converting compute, models, data, delivery, and user participation into internal rewards, and giving those rewards continuous and explainable liquidity. Bancor Plus preserves reserve-driven conversion, then adds participation funding, profit distribution, injection control, and audit observability. It is not a general-purpose exchange, stablecoin, fiat gateway, or any particular chain or ledger. It is an organizational settlement and incentive algorithm with explicit boundaries.

Research question: when enterprises and teams continuously issue AI-contribution rewards, how can they preserve exit liquidity and reward team members and users from real profit without fabricating trades, misclassifying capital funding, or allowing unbounded support?

2. Origin and Core Ideas of Bancor

2.1 Name and Motivation

The name Bancor honors John Maynard Keynes's wartime proposal for a supranational clearing currency. The original Bancor paper borrowed the name for a different problem: liquidity and conversion among digital tokens. It framed the double coincidence of wants as a barrier for long-tail currencies and proposed smart tokens whose prices are continuously computed from reserve state. Smart contracts were an early execution medium, not a prerequisite of the formulas.[1]

2.2 Constant Reserve Ratio and Continuous Pricing

The original model expresses instantaneous price using reserve balance R , smart-token supply S , and a constant reserve ratio. The price is recomputed after each purchase or liquidation, so price discovery unfolds asynchronously rather than requiring a matched bid and ask.[1]

$$P = R / (S * CRR)$$

$$\text{Purchase: } \Delta S = S * ((1 + \Delta R / R)^{CRR} - 1)$$

$$\text{Liquidation: } \Delta R = R * (1 - (1 - \Delta S / S)^{(1 / CRR)})$$

Effective execution integrates across infinitesimal state changes. A larger trade therefore moves farther along the curve and incurs more slippage, while deeper reserves reduce the movement produced by the same trade size.

2.3 Two-Reserve Conversion

A converter that holds two reserve assets can be viewed as entering an intermediate supply and exiting the other reserve. In the symmetric two-reserve case, the exact-input transformation has a compact direct form. For input reserve R_{in} , output reserve R_{out} , net input q , and output z :

$$z = q * R_{out} / (R_{in} + q)$$

$$(R_{in} + q) * (R_{out} - z) = R_{in} * R_{out}$$

$$p_{\text{marginal}} = R_{\text{USD}} / R_{\text{AIC}}$$

This transformation preserves the reserve product and is a two-asset constant-function market. Modern CFMM research formalizes these mechanisms through reserve-dependent trading functions and studies their pricing, reserve bounds, and path properties.[2][3]

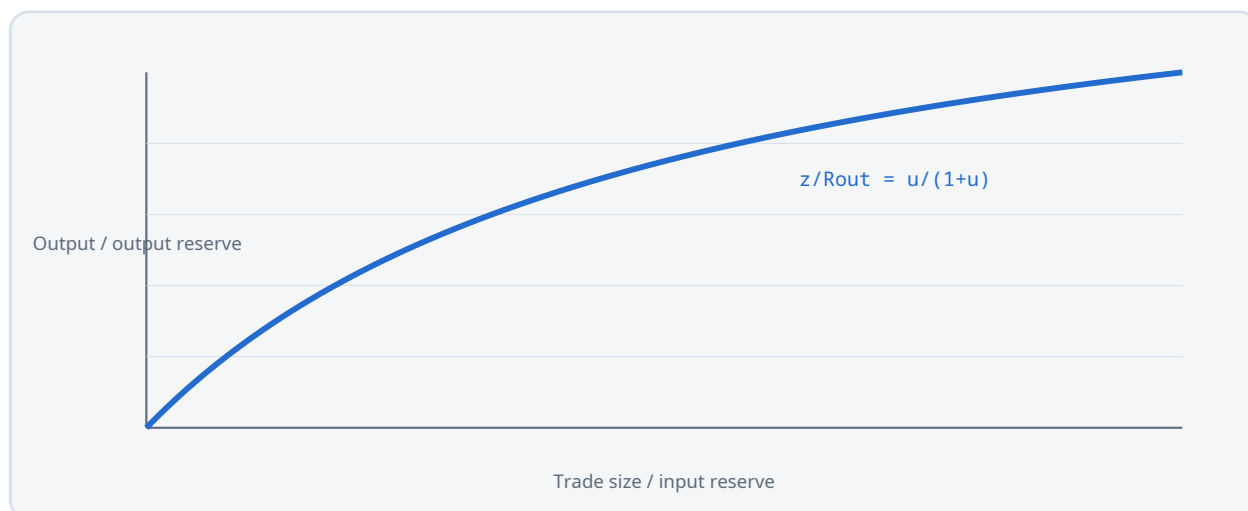


Figure 1. Normalized exact-input curve. Output keeps increasing, but marginal output declines with trade size.

3. Separating the Bancor Algorithm from Its Execution Medium

3.1 Algorithm Layer: State Transitions Only

Bancor is first a reserve-pricing and state-transition algorithm. It reads reserve balances, supply or paired-reserve state, policy parameters, and a conversion input; it then computes a quote, output, price impact, and next state. Given the same state, parameters, and input, it should produce the same result. This definition is independent of whether the state is held in a database, smart contract, blockchain, or another ledger.

Algorithm element	Content	Outside its scope
State	Reserve balances, supply, fees, and control parameters	Where state is stored and who custodies it
Input	Direction, amount, and minimum-output constraints	Identity, authentication, and authorization
Computation	Continuous price, output, impact, and next reserves	Signature format, confirmation, and commit protocol
Constraints	Nonnegative reserves, bounded output, deterministic result	Audit medium, disaster recovery, and upgrade governance

3.2 Execution and Settlement Layer

The execution layer turns an algorithmic result into an authoritative state change. It is responsible for custody, identity and permission checks, concurrency control, atomic commit, rollback, durable storage, receipts, audit, and final settlement. Different organizations may choose different carriers without changing the Bancor formulas.

Execution medium	Primary strengths	Costs and trust boundary
Authoritative server and database	Low latency, mature transactions, controlled recovery	Operator controls service, keys, and database
Permissioned ledger	Shared governance among known parties	Membership, consensus, and operations remain governed
Public chain	Public state verification and composability	Fees, confirmation delay, public ordering, and contract risk
Hybrid	Private execution with selected proofs or settlement published externally	Requires explicit reconciliation and finality rules

3.3 On-Chain Implementation: One Optional Deployment

When Bancor is implemented on a blockchain, the design must additionally address contract custody, transaction fees, confirmation and reorganization rules, public transaction ordering, front-running or extractable value, oracle dependencies, contract upgrades, and cross-chain risk. These are properties of the chosen on-chain execution environment. They must be analyzed separately rather than written into the definition of the reserve algorithm itself.

3.4 Current Bancor Plus Implementation Choice

The current Bancor Plus system uses an authoritative service, signed authorization, and atomic database transactions. This is an execution-layer choice, not a claim that Bancor requires centralized settlement. A future permissioned-ledger, public-chain, or hybrid carrier can preserve the same quote, funding-control, and profit-allocation algorithms if their state transitions and invariants remain equivalent.

Boundary rule: the algorithm explains how to calculate; the execution layer explains who may execute, where state lives, and how results are confirmed and audited. On-chain analysis applies only after a blockchain has been chosen as the execution medium.

4. Scope and Distinctive Features of Bancor Plus

Bancor Plus retains computable reserve-curve exchange while adding the funding, profit-distribution, control, and observation layers needed by enterprises and teams. Its distinctive contribution is not a replacement of the base trading equation, but a strictly accounted USD funding cycle connected to verifiable team and user contributions.

Dimension	Original Bancor idea	Current Bancor Plus implementation
Liquidity source	Algorithm reads reserve state	AIC/USD reserves plus a separate participation fund
Price formation	Reserve and CRR-based continuous pricing	Two-reserve exact-input curve; marginal price equals USD/AIC reserve ratio

Dimension	Original Bancor idea	Current Bancor Plus implementation
Additional funding	Initial reserves and purchase flows	First qualified AI-node participation forms budget; a controller later injects it
Profit and rewards	Outside the original conversion formula	Real operating profit is allocated to liquidity, teams, users, and retained earnings
Injection semantics	Not a core conversion action	Adds only USD reserve; moves no AIC and creates no trade or fee
Observability	Conversion and reserve state	Immutable receipts, asset flows, market versions, and funding-aware price candles
Settlement boundary	Defined by a separate execution layer	Current implementation uses an authoritative ledger, signed requests, and atomic database transactions

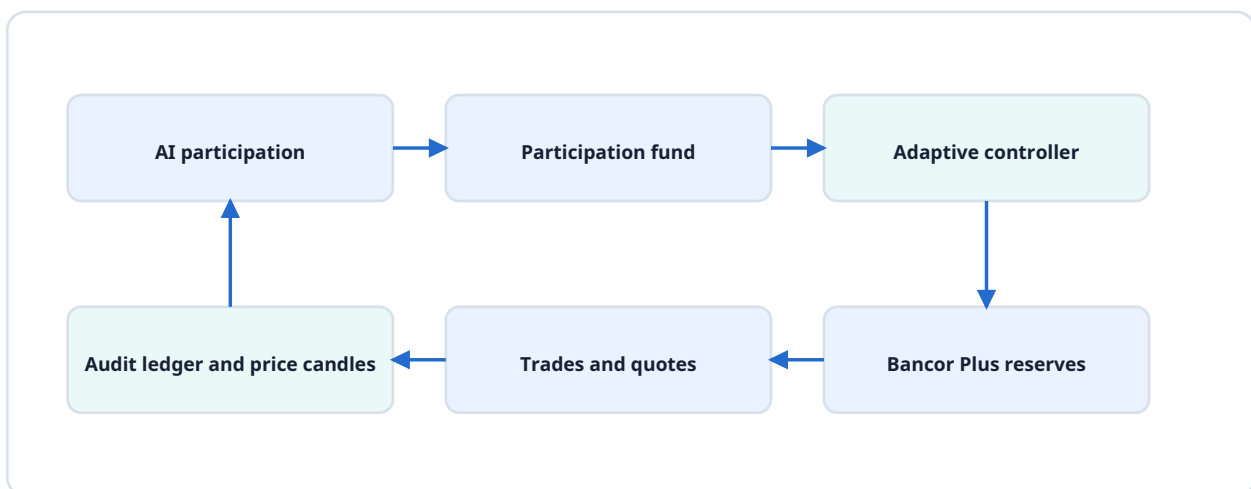


Figure 2. Bancor Plus funding and observation loop. Trade and non-trade funding paths remain separate in the ledger.

5. Trading Curve and Fees

5.1 Exact-Input Quote

For gross input Q and fee rate f in basis points, the curve receives the net amount q after fees. Output z depends on the two current reserves and the net input. Larger inputs move farther along the curve and create greater price impact.[4]

```

fee = Q * f / 10,000
q = Q - fee
z = q * Rout / (Rin + q)
price_impact_bps = q * 10,000 / (Rin + q)

```

5.2 Reserve Updates

Direction	User input	User output	Reserve transition
Buy AIC	Q USD	z AIC	USD reserve + q; AIC reserve - z
Sell AIC	Q AIC	z USD	AIC reserve + q; USD reserve - z

Fees are recorded outside curve reserves. A sell is additionally constrained by a hard USD reserve floor and an automatic pause threshold. Each quote binds the market version, input, output, minimum output, fee, nonce, and expiry. Execution recalculates the quote and rejects a higher fee or insufficient output.

6. Participation Fund for Enterprises and Teams

In the current implementation, an organization treats an allowlisted AI node as a qualified participation unit and separates node identity H from its asset account. First participation is accepted only after a valid asset-account binding with both owner and hardware signatures. Each node identity forms a 100 USD budget in a separate participation fund only once. Rebinding, account replacement, and replay do not form budget again.[5]

This USD does not enter the user's account and does not immediately enter Bancor reserves. It first forms an auditable fund F, separating budget formation as enterprise or team AI participation expands from the decision of when the market needs more USD depth. Node participation is the current implementation trigger; audited project budgets or settled operating profit could later replenish the fund, but unrealized pool appreciation must never be classified as profit. Budget formation and subsequent transfers appear in the asset ledger.

7. Adaptive One-Sided USD Injection

7.1 Variables and Target

Symbol	Meaning
B	Bancor USD reserve before the step
F	USD balance of the organizational participation fund
S	Target share of Bancor USD within B+F
d	Balance buffer around the target
r_min, r_max	Lower and upper fractions of F selectable per step
c	Per-step cap relative to the current Bancor USD reserve
A	Actual USD reserve addition

7.2 Algorithm

```

T = (B + F) * S
G = max(T - B, 0)
D = T * d
e = 0,                                if G <= D
e = min(1, (G - D) / (T - D)),        otherwise
r = r_min + (r_max - r_min) * e
R = F * r
C = B * c
A = min(F, R, G, C)

```

If the fund is empty, B is at or above target, or the gap lies inside the balance buffer, the step transfers zero while still allowing a decision receipt and the schedule to advance.

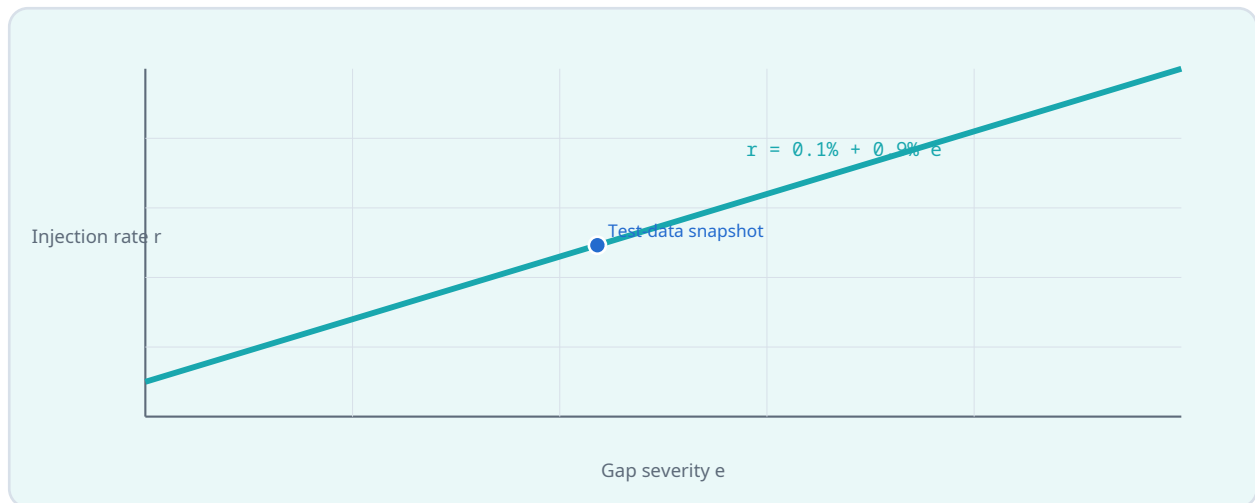


Figure 3. Under the current policy, the selected rate rises linearly from 0.1% to 1% as gap severity moves from 0 to 1.

7.3 Current Policy

Parameter	Test value	Purpose
Interval	600 seconds	Wait one full interval after the first eligible heartbeat; catch up at most once
Minimum rate	0.1%	Slowest extraction from F outside the buffer
Maximum rate	1%	Extraction rate at maximum gap severity
Target share	50%	Targets roughly equal USD balances in Bancor and the participation fund
Balance buffer	1%	Suppresses small adjustments near target
Pool step cap	2%	Limits one addition relative to current Bancor USD reserve

7.4 Market State After Funding

$$\begin{aligned}
 F' &= F - A \\
 B' &= B + A \\
 X' &= X && \text{(AIC reserve unchanged)} \\
 p' &= (B + A) / X \\
 k' &= X * (B + A) > X * B && \text{(when } A > 0)
 \end{aligned}$$

This is direct reserve funding, not a buy. It removes no AIC from the reserve and credits no user with AIC. It creates no trade identifier, trading volume, fee, or user-balance change. Because AIC reserve stays fixed while USD reserve grows, the marginal AIC price and available USD sell-side depth both increase. This economic effect differs from a neutral two-sided liquidity deposit.

8. Profit Distribution and Team/User Rewards

8.1 Define Distributable Profit First

Bancor supplies pricing, conversion, and liquidity; it does not manufacture operating profit. Profit must come from settled, reconcilable economic activity. For enterprises and teams, eligible sources may include AI inference, training, model hosting, data services, software subscriptions, and collected trading fees. A participation-fund transfer is an internal capital movement, not revenue. A higher marginal price after funding is unrealized valuation change, not distributable profit.

Item	Profit eligible?	Treatment
Settled AI-service revenue	Yes	Recognize after direct costs, refunds, taxes, and contract liabilities
Collected trading fees	Yes	Book as revenue or risk reserve under published policy
Participation-fund transfer	No	Internal capital movement, not income
Reserve appreciation	No	Unrealized value change, not a reward source
First-participation budget	No	Restricted budget held for its stated organizational purpose

```

Pi_t = max(R_t - C_t - L_t - V_t, 0)
R_t: settled operating revenue and eligible fees
C_t: verified operating costs
L_t: refunds, taxes, and contractual liabilities
V_t: required risk-reserve top-up

```

Only a positive Pi_t enters the reward waterfall. A loss period must not mint rewards or drain Bancor reserves to simulate profit; unrecovered losses and reserve deficiencies carry into the next settlement epoch.

8.2 Adaptive Profit Waterfall

Each settlement epoch first measures the Bancor USD reserve gap and selects α_t , the fraction of profit assigned to liquidity. A larger gap moves α_t toward its ceiling; a healthy market returns it toward its floor. This share enters the participation fund and remains subject to the controller in Section 7 rather than bypassing injection limits. The remaining profit is allocated to teams, users, and retained earnings through published governance weights.

```

h_t = clamp((T - B) / T, 0, 1)
alpha_t = alpha_min + (alpha_max - alpha_min) * h_t
Z = w_team + w_user + w_retain
Liquidity_t = alpha_t * Pi_t
Team_t = (1 - alpha_t) * w_team / Z * Pi_t
User_t = (1 - alpha_t) * w_user / Z * Pi_t
Retained_t = (1 - alpha_t) * w_retain / Z * Pi_t

```

The alpha bounds, governance weights, and per-epoch payout cap must be published and versioned before the epoch begins. This favors exit liquidity during stress while preventing managers from changing team or user shares after seeing the result.

8.3 Allocation Within Team and User Pools

Team rewards should reflect verifiable delivery, quality, reliability, reusable knowledge, and collaboration rather than hours or holdings alone. User rewards may reflect settled usage value, useful feedback, retention, and ecosystem contribution, while excluding self-trading, artificial volume, refunded orders, and unverifiable referrals. A square-root transform limits dominance by a single large contributor while preserving positive incentives.

```
s_i = max(weighted_verified_contribution_i - penalties_i, 0)
a_i = sqrt(s_i)
team_reward_i = Team_t * a_i / sum(a_j)
u_k = sqrt(max(verified_user_value_k - penalties_k, 0))
user_reward_k = User_t * u_k / sum(u_m)
```

8.4 Delivery Choices and Their Curve Effects

Delivery	Bancor effect	Use case
Transfer treasury AIC	No curve trade; organizational AIC inventory falls	Direct internal-asset rewards
Pay USD	No curve trade; distributable cash falls	Users need a stable accounting unit
Buy AIC and distribute	Real buy, fee, and price impact	Rewards should coincide with market demand

A buyback-and-distribute policy should use versioned quotes, cap each buy relative to Bancor USD reserves and maximum price impact, and execute across multiple windows. The profit receipt must trace to acquired AIC and final team/user recipients. An incomplete buyback returns to the pending-profit account and cannot be reported as a paid reward.

Recommended sequence: recognize real profit; allocate liquidity, team, user, and retained shares; then choose a direct payout or controlled buyback path. Profit recognition, liquidity funding, and reward delivery are linked but separately accounted actions.

9. Stability and Convergence

With no trades, no newly formed qualified-participation budget, and only internal F-to-B transfers, B+F remains constant, so target T is constant. Every positive transfer reduces gap G by exactly A. As long as the fund remains nonzero and the gap is outside the buffer, the sequence approaches the target monotonically. The rate ceiling slows fund depletion, the pool cap bounds relative market movement, and the balance buffer suppresses micro-adjustments near target.

```
B_next = B + A
F_next = F - A
T_next = (B_next + F_next) * S = T
G_next = max(G - A, 0)
```

The controller is one-way: it can move USD into Bancor but does not withdraw reserves when B exceeds target. An above-target state caused by trading only pauses new funding. The controller therefore supplies a bounded and auditable replenishment path, not a two-way peg or a fixed-price guarantee.

10. Numerical Case: Test-Data Snapshot

10.1 Test Reserves and Funding Step

The test-data snapshot uses AIC reserve $X = 155,725,642.07$, Bancor USD reserve $B = 36,573.93$, activation-fund balance $F = 94,509.59$, marginal price near 0.00023486 USD/AIC, and a 50 basis-point fee.[7]

Quantity	Result
Target reserve T	65,541.76 USD
Target gap G	28,967.83 USD
Buffer amount D	655.42 USD
Gap severity e	0.4363
Selected rate r	0.4927%
Rate amount R	465.65 USD
Pool step cap C	731.48 USD
Actual addition A	465.65 USD
Post-funding marginal price	about 0.00023785 USD/AIC, up about 1.2731%

In the same test-data snapshot, a gross 100 USD AIC purchase pays a 0.50 USD fee, sends 99.50 USD into the curve, and quotes approximately 422,504.82 AIC. Actual execution still requires the versioned quote, minimum-output check, and valid signature.

10.2 Enterprise Profit-Distribution Example

The following illustrative epoch is neither live revenue nor an adopted allocation policy. Suppose settled AI-business revenue and eligible fees total 120,000 USD, verified costs are 60,000 USD, refunds, taxes, and contract liabilities are 10,000 USD, and the risk-reserve top-up is 5,000 USD. Distributable profit Pi_t is 45,000 USD. With profit-liquidity bounds of 10% and 30%, the current reserve gap gives $h_t = 0.4420$ and $\alpha_t = 18.8395\%$. Remaining governance weights are team 5, user 3, and retained 2.

Purpose	Amount	Next action
Liquidity share	8,477.78 USD	Enter the participation fund; controller still limits injection
Team reward pool	18,261.11 USD	Allocate by square-root-adjusted team contribution scores
User reward pool	10,956.67 USD	Allocate by verifiable user-value scores
Retained earnings	7,304.44 USD	Keep for future operations and risk
Total	45,000.00 USD	Fully reconciles to distributable profit

This step assigns purposes; it does not prove payment. Team and user pools still require recipient-level records. If rewards use an AIC buyback, Section 8.4's controlled execution applies and the final AIC received must reconcile to distribution receipts.

11. Auditability, Price Candles, and Safety Boundaries

11.1 Separate Observation of Non-Trade Funding

Each funding step atomically commits the fund debit, Bancor USD reserve increase, market-version update, immutable funding receipt, asset ledger, and explorer flow. Any failure rolls back the entire step. Funding events enter pool-marginal-price candles so an interval with no trade but a changed reserve still shows the real price state. Funding count and USD amount remain separate and do not enter trade count, AIC/USD volume, or fee totals.[6]

11.2 Controls

Sells are constrained by a hard USD floor and an automatic pause threshold.

Quotes bind a market version and are recalculated at execution.

Signatures bind direction, amount, fee, minimum output, nonce, task, and expiry.

Account, market, ledger, fee, and trade rows update in one immediate transaction.

Funding receipts and asset flows are immutable and subject to offline reconciliation.

11.3 Remaining Limitations

The current database, system keys, and settlement service remain critical control points; this is execution-layer risk, not reserve-curve formula risk.

Internal USD is neither a bank deposit nor a fiat redemption right; AIC carries no capital or price-stability guarantee.

One-sided funding raises marginal price and changes the curve invariant, so it is an economically consequential policy action.

The controller injects but does not withdraw. It can only stop when external trades move reserves above target.

Targets, rates, and caps are governance parameters rather than automatically optimal risk estimates; they require stress testing.

A single authoritative Core and database provide atomic consistency but still require future work for multi-writer and regional failover.

12. Conclusion

Bancor's central contribution was to turn liquidity from waiting for a counterparty into interacting with reserves and a formula. Bancor Plus applies that idea to enterprises and teams in the AI era. Qualified participation forms separate budget; the adaptive algorithm selects a funding rate from the reserve gap and constrains each transfer. Real operating profit then follows a separate waterfall into liquidity, team, user, and retained shares. Trade, funding, and rewards remain distinct in accounting and audit.

The design should not be evaluated as a promise to eliminate volatility. Its value is a computable, configurable, rollback-safe, and auditable way to connect AI-business revenue, internal rewards, and exit liquidity. Future work should study profit-recognition policy, manipulation resistance of contribution scores, fund lifetime under extreme selling, two-way rebalancing, governance, multi-node consistency, and public verification.

References

- [1] E. Hertzog, G. Benartzi, and G. Benartzi. Bancor Protocol: Continuous Liquidity and Asynchronous Price Discovery for Tokens through their Smart Contracts. Draft 0.77, 2017. <https://resources.bancor.network/pages/BancorProtocolWhitepaper.pdf>
- [2] G. Angeris and T. Chitra. Improved Price Oracles: Constant Function Market Makers. Proceedings of AFT 2020. <https://arxiv.org/abs/2003.10001>
- [3] G. Angeris, A. Agrawal, A. Evans, T. Chitra, and S. Boyd. Constant Function Market Makers: Multi-Asset Trades via Convex Optimization. 2021. <https://arxiv.org/abs/2107.12484>
- [4] NNI implementation reference. `nni_server/bancor_math.mjs` and `nni_server/storage.mjs`, commit `f21375395e0204eec7f42f6c54af9a9bd5c16210`, reviewed 2026-09-28.
- [5] NNI implementation reference. `nni_server/activation_fund.mjs`, one-time 100 USD qualified-participation budget policy, reviewed 2026-09-28.
- [6] NNI implementation reference. `nni_server/activation_fund_liquidity.mjs`, `activation_fund_liquidity_config.mjs`, and pool-price candle projections, commit `f21375395e0204eec7f42f6c54af9a9bd5c16210`.
- [7] Test-data source endpoint. <https://api-1.matrixai.one/v1/nni/server/bancor/market>, accessed 2026-09-28 08:30:31 CST.

Reproducibility note: all Bancor Plus numerical results were recomputed with the current formulas. The test-data snapshot is illustrative and is not an earnings forecast, valuation opinion, or trading recommendation.